

Leiden Conventions for Greek Numismatic Epigraphy

Chris Hopkins, DRAFT 18 Oct 2017

Epigraphers have long dealt with the need to document inscriptions in a comprehensible fashion. A measure of standardization was achieved with development of conventions by scholars of the Union Académique Internationale meeting at the University of Leiden in 1931.¹ These conventions established rules to indicate the condition of an epigraphic or papyrological text in modern edition. They are simple and straightforward but often ignored in numismatic documentation.

This paper draws upon the traditions of numismatists and recommends the sigla used to achieve standardized documentation of inscriptions. It follows the Leiden Conventions with several exceptions widely used in numismatic publications and considers the different requirements of plain text and decorated (rich) text word processors.

For more than 45 years David Sellwood's *An Introduction to the Coinage of Parthia* has been the preeminent reference for cataloging Parthian coins. I verified the near-universality of Sellwood as the post-1971 standard reference when populating a large database of Parthian coins, drawing from the catalogs of more than 2,000 sources. In the process I also confirmed the great divergence in techniques used to document Greek numismatic inscriptions.

A distinctive feature of Sellwood's *Introduction* is his handwritten text. The calligraphy is not only beautiful in its own right, it allowed flexibility in spacing, hand drawn devices and monograms that capture minor differences, and faithful reproduction of the inscriptions. No typesetter, even if equipped with an extensive array of computer fonts, could hope to match the accuracy of his text. Both his 1971 and 1980 editions are composed entirely of text, figures, maps, and tables drawn by hand. The only exceptions are the photographic plates that were, characteristically, created using the traditional plaster casts. In a world where technology is moving quickly – and where many students are no longer taught cursive handwriting – Sellwood's work provides the endpoint to an earlier and more engaging era in numismatic publishing.

His methodology had one great advantage: as shown in figure 1, he illustrates Greek inscriptions as they appear on the coins, usually in the square arrangement so dominant on Parthian coins.²

¹ McLean, p. 27

² Sellwood (1980), p. 181



Fig 1

The graphic documentation of the inscription plus a cartoon of the design leaves no question as to the content and physical layout. While suitable for printed text, the extensive use of illustrations is rarely practical in digital documentation, particularly where large numbers of inscriptions must be entered. How then does a scholar or cataloger reliably indicate not only the inscription, but its arrangement? How does one indicate what is missing and what is erroneously present?

Digital plain text is devoid of decoration and is typically entered into a simple text editor or a database field. While Unicode fonts offer many accented and special characters, the decoration of text with color, underline, overline, italics, superscripts and other features is generally the province of sophisticated presentation formats. Plain text editors can display only the Unicode characters, not complex combining characters and other Unicode features, and thus display no decoration. On the other hand, decorated text is easily achieved using sophisticated rich text word processors or the presentation formatting capabilities of hypertext mark-up languages.

Examine the inscription on the central coin image in figure 1. The first line below gives the *pro forma* inscription. Line b applies the Leiden Conventions sigla and the last line repeats line b using the ancient and archaic Greek glyphs of the Numismatica Pro or Cardo fonts³ to closely imitate letters on the coin. The regularity of inscriptions on Parthian coins allows restorations that otherwise might be marked as lacuna.

- a. ΒΑΣΙΛΕΩΣ ΒΑΣΙΛΕΩΝ ΑΡΣΑΚΟΥ ΕΥΕΡΓΕΤΟΥ ΑΥΤΟΚΡΑΤΟΡ ΕΠΙΦΑΝΟΥΣ ΦΙΛΕΛΛΗΝΟΣ
ΦΙΛΟΡΩΜΑΙΟΥ
- b. [ΒΑΣΙΛΕΩΣ] / ΒΑΣΙΛΕ[ΩΝ] // [ΑΡΣΑΚΟΥ / Ε]ΥΕΡΓΕΤΟΥ // [Α]ΥΤΟΚΡΑΤΟΡ // [Ε]ΠΙΦΑΝΟΥΣ
/ [ΦΙ]ΛΕΛΛΗΝΟΣ // ΦΙΛΟΡΩΜΑΙΟ[Υ]

³ Numismatica Pro font is available at <http://parthia.com/fonts>. David Perry's Cardo font is available at <http://scholarsfonts.net/cardofnt.html>

- c. [ΒΑΣΙΛΕΥΣ] / ΒΑΣΙΛΕ[ΛΝ] // [ΑΡΣΑΚΩΝ / Ε]ΥΕΡΓΕΤΩΝ // [Α]ΥΤΟΚΡΑΤΩΡ // [Ε]ΠΙ+ΑΝΩΝΣ / [+Ι]ΛΕΛΛΗΝΩΣ // +ΙΛΟΡΩΜΑΙΟ[Υ]

The following numbered sections describe the use of capital letters, the Leiden sigla used in plain text documentation, the few sigla that require decorated text, and lastly the non-Leiden arrows that occasionally have been used by numismatists. The preferred Unicode usage⁴ is given for each followed by acceptable alternates in order of preference.

1. Use capital letters to document Greek numismatic inscriptions.

ΑΒΓΔ Inscriptions in capital letters

Numismatic epigraphy has traditionally used capital letters. This is a significant exception to conformance with the Leiden Conventions; when used in a numismatic inscription, capital letters do not have the Leiden Conventions' meaning of "reading clear, interpretation unknown".

Do not use look-alike roman letters in digital documents. Use only capital letters in the Unicode Greek and Coptic range U+0370 to U+03FF, or alternate Greek letterforms located in the Unicode Private Use Areas of specialty fonts. Computer search and sort capabilities rely on use of the correct Unicode letters.

2. Sigla suitable for documenting inscriptions in either plain or decorated text:

[ΒΑΣ] Restorations of missing or damaged letters

Letters missing due to damage, or are off flan, or incompletely struck; the missing letters are restored by the editor.

Unicode U+005B Left Square Bracket and U+005C Right Square Bracket enclosing one or more letters.

— Spacing break

A dash is used to indicate the letters are not contiguous, as when the lettering is spaced around a device.

Unicode U+2014 Em Dash.

Alternate: U+2013 En Dash or a U+002D Hyphen-Minus.

[. . .] or [. . 12 . .] Lacuna (extent known)

⁴ The Unicode Standard, version 10.0 (2017), provides character charts for the listed codes

Lacuna or gap in the original inscription, not restored by the editor (extent known). The number of points always indicates the number of lost letters; an Arabic numeral may be used to indicate a large number of missing characters.

Unicode U+005B Left Square Bracket and U+005C Right Square Bracket enclosing one or more U+002E Full Stop. Use space between dots.

Alternate: U+00B7 Middle Dot instead of Full Stop.

[- - -] Lacuna (extent unknown)

Lacuna or gap in the original inscription, not restored by the editor (extent unknown); lost or illegible letters of an uncertain number. The number of dashes has no significance and does not suggest the number of missing letters.

Unicode U+005B Left Square Bracket and U+005C Right Square Bracket enclosing one or more Unicode U+2013 En Dash. Use space between dashes.

Alternate: U+002D Hyphen-Minus or U+2012 Figure Dash, or U+2014 Em Dash.

<TOY> Omitted characters restored: additions and substitutions

Characters erroneously omitted or left incomplete by the engraver that are restored or corrected by the editor. Note: The Less Than Sign operates as an escape character in HTML. When preparing text for presentation in web browsers, encode the < and > signs as < and > in the source to avoid problems.

Unicode U+003C Less Than Sign and U+003E Greater Than Sign enclosing one or more letters.

Alternate: U+2329 Left-Pointing Angle Bracket and U+232A Right-Pointing Angle Bracket.

ΔΥΣΤ(POY) Resolution of abbreviations, symbols and ligatures

Resolution of a symbol or abbreviation expanded by the editor; use uppercase letters to complete words abbreviated in the inscription.

Unicode U+0028 Left Parenthesis and U+0029 Right Parenthesis enclosing one or more Greek capital letters that complete the abbreviation.

ΔΙΚΑΙΟΥ (K retrograde) Explanation

Editor provides lowercase description to explain special aspects of the inscription.

Unicode U+0028 Left Parenthesis and U+0029 Right Parenthesis enclosing the explanation.

{N} Superfluous

Letters in the inscription considered erroneous and superfluous by the editor; letters added in error by the engraver and excised by the editor.

Unicode U+007B Left Curly Bracket and U+007D Right Curly Bracket enclosing one or more letters.

E/ Possible reading

Letters for which sufficient traces remain to print them in the text, but not enough to exclude other readings.

Letter immediately followed by Unicode U+002F Solidus.

[[ABΓ]] or [ABΓ] Rasures

Letters deliberately erased from die or coin in antiquity.

Two Unicode U+005B Left Square Bracket and two U+005C Right Square Bracket enclosing one or more letters.

Alternate: U+301A Left White Square Bracket and U+301B Right White Square Bracket.

/ (or |) Line or section separator

Use a single solidus mark or vertical bar to separate individual lines or groups of lines. While the Leiden Conventions specify the vertical line, the solidus is used more often for numismatic epigraphy due to the frequent appearance of vertical lines that represent various letters, especially in later Parthian inscriptions.

Unicode U+002F Solidus.

Alternate: Unicode U+007C Vertical Line.

// or || (or || or ||) Section separator

Use a pair of solidus marks, or a pair of vertical bars to separate groups of lines in an inscription; this is also used to separate the main inscription from letters in exergue or fields. While the Leiden Conventions specify the double vertical line, the solidus is used more often for numismatic epigraphy due to the frequent appearance of vertical lines to represent various letters, especially in later Parthian inscriptions.

Two Unicode U+002F Solidus or one U+2AFD Double Solidus Operator. Note the double solidus operator is found in only a few fonts.

Alternate: Two Unicode U+007C Vertical Line, or one Unicode U+2016 Double Vertical Line.

+++ Trace letters, not restorable

Traces of letters, insufficient for restoration.

One or more Unicode U+002B Plus Sign, with spaces between.

\ / Insertion

Insertion above the line.

Unicode U+005C Reverse Solidus and U+002F Solidus enclose the letters that were inserted above the line on the original inscription.

vac. or vacat Space left empty

Latin *vacat*: "I am empty"; space left empty on the coin or where the remainder of the line has not been engraved.

3. Sigla suitable for documenting inscriptions in plain and decorated text: The following are included for completeness but rarely have been used in numismatics.

ΒΑΣΙΛΕΩΝ Parts read earlier now missing (underlined)

Underline letters in an inscription that have been broken or further damaged since first read and published, and thus no longer legible. A possible scenario in numismatics is legibility lost due to corrosion.

To each underlined letter add a Unicode U+0332 Combining Low Line. The complexity of using Unicode combining glyphs with Greek letters exceeds the capability of a plain text Unicode document.

Alternate: Most word processors, and markup languages such as HTML, are capable of displaying underlines.

ΒΑΣΙΛΕΩΝ Ambiguous characters (underdotted)

Use dots under letters damaged or otherwise unclear in the inscription or ambiguous outside of their context; the character is in evidence on the support but its exact identity is not clear. Use underlining for an inscription broken or further damaged since first read and published, and thus no longer legible.

Unicode U+0323 Combining Dot Below. The complexity of using Unicode combining glyphs with Greek letters exceeds the capability of a plain text Unicode document. Most word processors such as Microsoft Word, and markup languages such as HTML, are capable of displaying underdots.

4. Non-Leiden sigla:

← ↑ → ↓ ↻ ↺ *et cetera* **Directional Arrows**

Some editors use arrow glyphs to indicate directional aspects of inscriptions. Arrows can indicate reading from the inside or outside, the text direction of Boustephedon lines, a clockwise circular legend, etc.

Unicode contains many arrow characters useful to graphically explain inscriptions. These are contained in Unicode ranges U+2190 – U+21FF Arrows Range, U+27F0 – U+27FF Supplemental Arrows-A Range and U+2900 – U+297F Supplemental Arrows-B Range.

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A personal note: I came to know David Sellwood through his publications and eventually had the privilege of meeting him in 2000. A generous and wonderful man, he invited me to his home on several occasions. A highlight was the opportunity to photograph his superb collection and audiotape his commentary about each Parthian coin.

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